



Suggested Solutions for 2016 H2 Chemistry A Level Papers

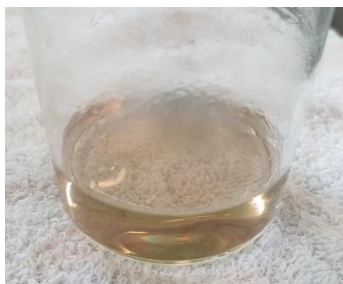
Paper 1

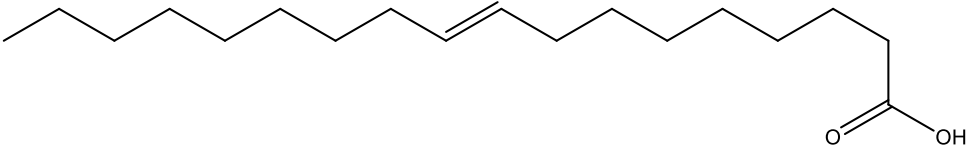
1	B	11	C	21	B	31	B
2	D	12	A	22	B	32	A
3	C	13	C	23	B	33	C
4	D	14	B	24	A	34	D
5	A	15	D	25	D	35	C
6	C	16	A	26	C	36	B
7	C	17	A	27	A	37	C
8	D	18	A	28	B	38	B
9	B	19	D	29	B	39	A
10	D	20	D	30	B	40	C

Paper 2

1	(a)	- Solid sample of azurite dissolves in sulfuric acid to form a blue solution - Effervescence of colourless odourless gas (that gives a white precipitate with limewater)	[1]
	(b)	<u>Calculations of suitable mass of powdered rock</u> Amount of sulfuric acid that reacted with azurite $= 0.75 \times \frac{50.00}{1000} \times 1.00 = 0.0375 \text{ mol}$ Amount of azurite that reacted = $\frac{0.0375}{3} = 0.0125 \text{ mol}$ Mass of azurite that reacted = $0.0125 \times 344.5 = 4.31 \text{ g}$ Suitable mass of powdered rock = $\frac{100}{90} \times 4.31 = 4.78 \text{ g}$ <u>Pre-calculations</u> Assuming an expected titre volume of 25.00 cm^3 Amount of NaOH added = $\frac{25.00}{1000} \times 0.100 = 0.00250 \text{ mol}$ Amount of sulfuric acid reacted = $\frac{0.00250}{2} = 0.00125 \text{ mol}$	[11]

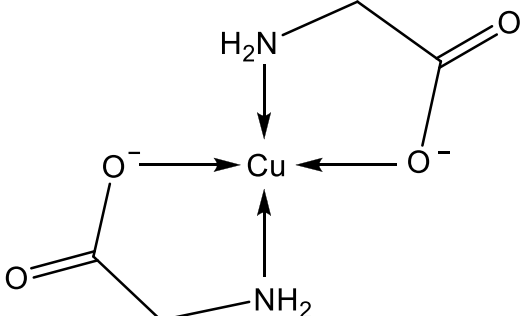
	Since amount of sulfuric acid remaining after reaction with azurite $= 0.25 \times \frac{50.00}{1000} \times 1.00 = 0.0125 \text{ mol}$ A 25.0 cm³ aliquot of sulfuric acid diluted to 250cm³ will contain the required 0.00125 mol of sulfuric acid. <u>Practical Details</u> 1. Weigh accurately about 4.78 g of powdered rock a <u>clean dry weighing bottle</u>. Record the mass of weighing bottle and the powdered rock. 2. Transfer the powdered rock into the 50.00 cm³ sulfuric acid in a 250 cm³ conical flask. 3. Swirl and allow the reaction to go to completion <u>until no more effervescence is observed</u>. 4. Reweigh the weighing bottle and record the mass of weighing bottle and the residue. 5. Transfer the solution in the 250 cm³ conical flask into a 250 cm³ volumetric flask. 6. Rinse the conical flasks with deionised water several times and transfer the rinsings into the same volumetric flask. 7. Make-up to the mark with deionised water. Stopper the flask and <u>shake it well to obtain a homogenous solution</u>. Label this solution FA1. 8. Pipette 25.0 cm³ of FA1 into a 250 cm³ conical flask. Add 2 drops of phenolphthalein indicator solution. 9. Titrate the contents of the conical flask with 0.100 mol dm⁻³ NaOH from a 50 cm³ burette, adding NaOH(aq) <u>dropwise near the end-point</u>, until the colour of the solution turns from <u>light blue to light purple</u>. 10. Repeat the titration as many times as necessary to obtain two consistent titres <u>within 0.10 cm³ in difference</u>. Mass of the powdered rock = mass of weighing bottle and powdered rock - mass of the weighing bottle and residue = P Let the average titre value (of two titres within 0.10 cm³) be E cm³. Amount of NaOH added = $\frac{E}{1000} \times 0.100 = (1 \times 10^{-4})\mathbf{E} \text{ mol}$ Amount of sulfuric acid reacted = $(1 \times 10^{-4})\mathbf{E} \div 2 = (5 \times 10^{-5})\mathbf{E} \text{ mol}$ Amount of excess sulfuric acid in the reaction mixture $= (5 \times 10^{-5})\mathbf{E} \times \frac{250}{25.0} = (5 \times 10^{-4})\mathbf{E} \text{ mol}$ Amount of sulfuric acid that reacted with azurite $= \frac{50.00}{1000} \times 1.00 - (5 \times 10^{-4})\mathbf{E} = (0.05 - (5 \times 10^{-4})\mathbf{E}) \text{ mol}$ Amount of pure azurite present in powdered rock	
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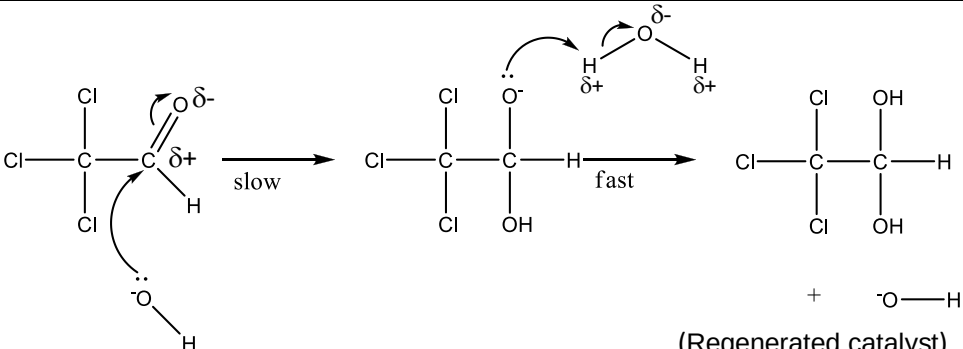
		$= \frac{1}{3} (0.05 - (5 \times 10^{-4})\mathbf{E}) \text{ mol}$ % by mass of pure azurite present in powdered rock sample $= \left[\frac{1}{3} (0.05 - (5 \times 10^{-4})\mathbf{E}) \times 344.5 \div \mathbf{P} \right] \times 100\%$ Note: If methyl orange indicator is used, With the acidic solution (with Cu^{2+} ions) in the conical flask, colour change at end point is from purple to green. With the alkaline solution in the conical flask, colour change at end point is from green to brown. 		
2	(a)	(i)	More energy is required to overcome the stronger hydrogen bonds between the fatty acid molecules compared to the weaker permanent dipole-permanent dipole interaction between the triester molecules.	[2]
		(ii)	Stearic acid has a larger number of electrons compared to palmitic acid. More energy is required to overcome the stronger instantaneous dipole–induced dipole (id-id) interactions between stearic acid molecules than the id-id interactions between palmitic acid.	[2]
		(iii)	The more C=C are present, the lower the melting point of the acid.	[1]
	(b)	(i)	Electrophilic addition	[1]
		(ii)	Mass of Iodine = $\frac{0.237}{0.256} \times 100 = 92.6 \text{ g}$	[1]
		(iii)	For 100g sample of olive oil: Amount of oil = $\frac{100}{782} = 0.128 \text{ mol}$ Amount of $\text{I}_2 = \frac{92.6}{(2)(127)} = 0.364 \text{ mol}$ Average no of C=C = $\frac{0.364}{0.128} = 2.85$	[3]
		(iv)	Since each oleic acid has 1 C=C, the triesters in vegetable oil should have 3 C=C. The average number of C=C calculated in (iii) suggests that a <u>small amount of vegetable oil are made up of only 2 oleic acid with either a palmitic or stearic acid molecule.</u>	[1]

	(c)	(i)	H ₂ (g), Ni catalyst, heat	[2]
		(ii)		[1]
	(d)	The functional groups that react are <u>esters</u> and <u>alkene</u>. The type of reactions are <u>oxidation</u> of alkene and <u>hydrolysis</u> of esters. <u>Oxidation of alkene</u> $ \begin{array}{c} \text{H} \\ \diagdown \\ \text{C} \\ \diagup \\ \text{R}_1 \end{array} = \begin{array}{c} \text{R}_2 \\ \diagup \\ \text{C} \\ \diagdown \\ \text{H} \end{array} + 4[\text{O}] \longrightarrow \begin{array}{c} \text{R}_1 \\ \\ \text{O}=\text{C}-\text{OH} \end{array} + \begin{array}{c} \text{R}_2 \\ \\ \text{O}=\text{C}-\text{OH} \end{array} $ <u>Hydrolysis of esters</u> $ \begin{array}{c} \text{R}_1 \\ \\ \text{C} \\ \\ \text{O} \end{array} - \text{O} - \text{R}_2 + \text{H}_2\text{O} \longrightarrow \begin{array}{c} \text{R}_1 \\ \\ \text{C} - \text{OH} \\ \\ \text{O} \end{array} + \text{HO} - \text{R}_2 $		[4]

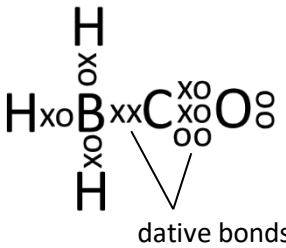
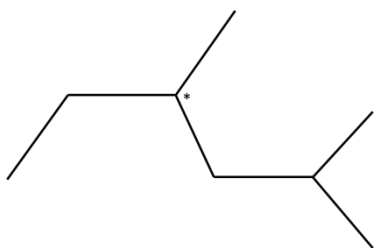
3	(a)	SiC has a giant covalent structure. A large amount of energy is required to overcome the strong covalent bonds between Si and C atoms. Hence, SiC has a high melting point and it is resistant to melting.		[3]																
	(b)	(i)	$C_xH_y + (x + y/4) O_2 \rightarrow x CO_2 + y/2 H_2O$	[1]																
		(ii)	Heterogeneous catalyst. Reactant particles diffuse to the surface of the catalyst and form weak interactions with the catalyst – <u>adsorption</u> . This will <u>weaken the covalent bonds</u> in the reactant molecules and at the same time bring the reactant molecules into <u>closer proximity</u> and into the <u>correct orientation</u> for reaction to occur. Hence, an <u>alternate reaction pathway with lower E_a</u> is provided. More reactant particles will possess <u>kinetic energy that is equal or greater than the lowered E_a</u> . <u>Frequency of effective collisions increase</u> which results in increase in rate of reaction. After reaction is complete, the product molecules <u>desorb</u> from the surface of the catalyst and diffuse away.	[4]																
	(c)	(i)	Oxidation state of nitrogen increases from -3 in NH_3 to 0 in N_2 . Oxidation state of nitrogen also decreases from $+4$ in NO_2 to 0 in N_2 .	[1]																
		(ii)	$8 NH_3 + 6 NO_2 \rightarrow 7 N_2 + 12 H_2O$	[1]																
	(d)	(i)	Since NO_2 and SO_2 are in the same gaseous phase, NO_2 acts as a <u>homogenous catalyst</u> in the oxidation of atmospheric SO_2 . Step 1: $NO_2 + SO_2 \rightarrow NO + SO_3$ Step 2: $NO + \frac{1}{2}O_2 \rightarrow NO_2$ Overall: $SO_2 + \frac{1}{2}O_2 \rightarrow SO_3$	[3]																
		(ii)	Atmospheric NO_2 results in the <u>formation of acid rain</u> that causes damage to buildings. NO_2 forms <u>photochemical smog</u> which causes respiratory problems. It also leads to the <u>excessive growth of algae</u> which depletes oxygen in water, affecting the marine ecosystem adversely.	[2]																
		(iii)	<table><tr><td></td><td>$2NO_2(g)$</td><td>$\rightleftharpoons$</td><td>$N_2O_4(g)$</td></tr><tr><td>Initial partial pressure / Pa</td><td>1.5×10^{-3}</td><td></td><td>0</td></tr><tr><td>Change in partial pressure / Pa</td><td>$-2x$</td><td></td><td>$+x$</td></tr><tr><td>Eqm partial pressure / Pa</td><td>$(1.5 \times 10^{-3}) - 2x$</td><td></td><td>$x$</td></tr></table> $K_p = \frac{P_{N_2O_4}}{(P_{NO_2})^2}$ $6.25 \times 10^{-5} = \frac{x}{[(1.5 \times 10^{-3}) - 2x]^2}$		$2NO_2(g)$	$\rightleftharpoons$	$N_2O_4(g)$	Initial partial pressure / Pa	1.5×10^{-3}		0	Change in partial pressure / Pa	$-2x$		$+x$	Eqm partial pressure / Pa	$(1.5 \times 10^{-3}) - 2x$		x	[2]
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Eqm partial pressure / Pa	$(1.5 \times 10^{-3}) - 2x$		x																	

			Assume x is small, so $(1.5 \times 10^{-3}) - 2x \approx 1.5 \times 10^{-3}$. $6.25 \times 10^{-5} = \frac{x}{(1.5 \times 10^{-3})^2}$ Partial pressure of $\text{N}_2\text{O}_4 = x = \underline{1.41 \times 10^{-10} \text{ Pa}}$	
4	(a)	(i)	K: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ Cu: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$	[1]
		(ii)	1st IE of K = 418 kJ mol^{-1} 1st IE of Cu = 745 kJ mol^{-1} Compared to K, Cu has a higher 1st IE. Cu has higher nuclear charge while shielding effect remains relatively constant. Thus, Cu has a higher effective nuclear charge and smaller atomic radius. This increases nuclear attraction on the valence electron, making it more difficult to be removed to form Cu^+. The lower reactivity of Cu is also supported by the more positive value of $E^\ominus(\text{Cu}^+/\text{Cu})$, as compared to $E^\ominus(\text{K}^+/\text{K})$, which reflects the lower tendency for copper to be oxidised (lower reducing power of copper). $\text{Cu}^+ + e^- \rightleftharpoons \text{Cu} \quad E^\ominus = +0.52 \text{ V}$ $\text{K}^+ + e^- \rightleftharpoons \text{K} \quad E^\ominus = -2.92 \text{ V}$	[4]
	(b)	(i)	With oxygen: $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4e^- \rightleftharpoons 2\text{H}_2\text{O}(\text{l}) \quad E^\ominus = +1.23 \text{ V}$ $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightleftharpoons \text{Cu}(\text{s}) \quad E^\ominus = +0.34 \text{ V}$ $2\text{Cu}(\text{s}) + 4\text{H}^+(\text{aq}) + \text{O}_2(\text{g}) \rightarrow 2\text{Cu}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$ $E^\ominus_{\text{cell}} = +1.23 - (+0.34) = +0.89 \text{ V} > 0 \text{ V}$ (reaction is spontaneous) Hence, Cu reacts with acid in the presence of oxygen. Without oxygen: $2\text{H}^+(\text{aq}) + 2e^- \rightleftharpoons \text{H}_2(\text{g}) \quad E^\ominus = 0.00 \text{ V}$ $\text{Cu}^{2+}(\text{aq}) + 2e^- \rightleftharpoons \text{Cu}(\text{s}) \quad E^\ominus = +0.34 \text{ V}$ $\text{Cu}(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ $E^\ominus_{\text{cell}} = 0.00 - (+0.34) = -0.34 \text{ V} < 0$ (reaction is non-spontaneous) Hence Cu(s) does not react with acid without oxygen.	[3]
		(ii)	The concentration of oxygen in air is low (Air is only 20% oxygen) so the frequency of effective collisions is low, leading to a low rate of corrosion.	[1]
		(iii)	Species responsible for dark blue solution: $[\text{Cu}(\text{NH}_3)_4]^{2+}$ $\text{O}_2 + 2\text{H}_2\text{O} + 4e^- \rightleftharpoons 4\text{OH}^-$ $[\text{Cu}(\text{NH}_3)_4]^{2+} + 2e^- \rightleftharpoons \text{Cu} + 4\text{NH}_3$ [R] $\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) + 4e^- \rightarrow 4\text{OH}^-(\text{aq})$ [O] $\text{Cu} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+} + 2e^- \quad \times 2$	[2]

			Overall equation: $2\text{Cu} + 8\text{NH}_3 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2[\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{OH}^-$	
	(c)	(i)	A ligand is an ion or molecule which contains at least one atom bearing a lone pair of electrons which can be donated into the low-lying vacant orbital of the central metal atom or ion forming a co-ordinate (dative) bond.	[1]
		(ii)		[1]

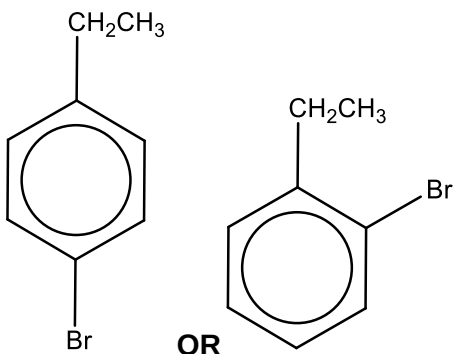
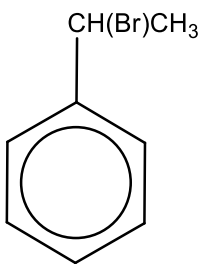
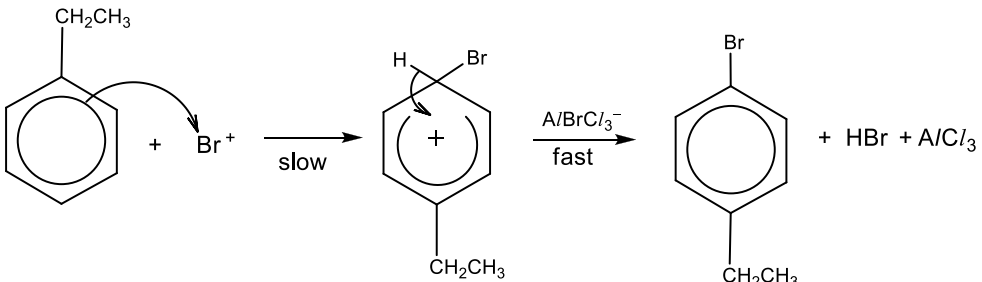
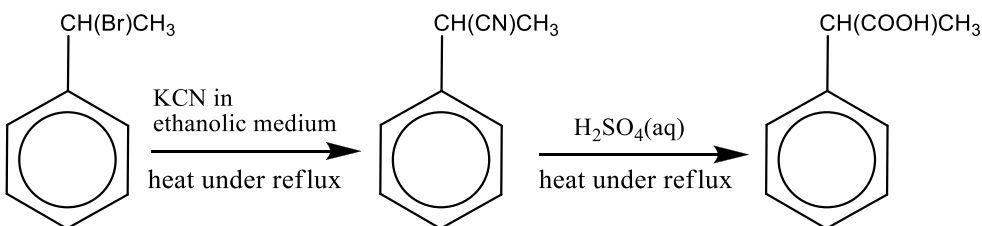
5	(a)	(i)	Reagent: 2, 4-DNPH Observations: <u>orange ppt</u> is observed if trichloroethanal is present, else solution will remain orange and <u>no ppt</u> is observed. Chloral hydrate will not give an orange ppt with 2,4-DNPH.	[2]
		(ii)	 (Regenerated catalyst)	[4]
		(iii)	$\text{CCl}_3\text{CHO} + \text{OH}^- \rightarrow \text{HCO}_2^- + \text{CHCl}_3$	[1]
	(b)	(i)	Trichloroethanoic acid is the stronger acid. The presence of electron-withdrawing Cl atoms disperses the negative charge on the COO^- group in $\text{CCl}_3\text{CO}_2^-$, making $\text{CCl}_3\text{CO}_2^-$ anions more stable than CH_3CO_2^- anions.	[3]
		(ii)	(Not in 9729 syllabus) At low pH, <u>$-\text{COO}^-$ on the R group of the amino acid residue becomes $-\text{COOH}$</u> while at high pH, <u>$-\text{NH}_3^+$ on the R group of the amino acid residue becomes $-\text{NH}_2$</u>. This disrupts the <u>ionic interactions</u> in the tertiary structure, and hence lead to denaturation. At low pH, <u>$-\text{NH}_2$ on the R group of the amino acid residue becomes $-\text{NH}_3^+$</u> while at high pH, <u>$-\text{COOH}$ on the amino acid residue becomes $-\text{COO}^-$</u>. This disrupts the <u>hydrogen bonds</u> in the tertiary structure, and hence leads to denaturation.	[2]

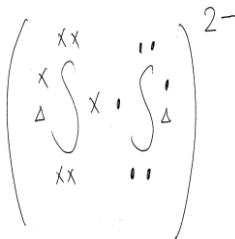
Paper 3

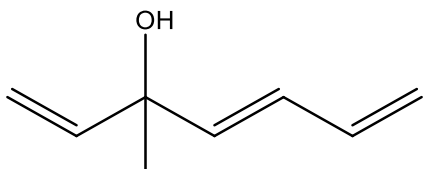
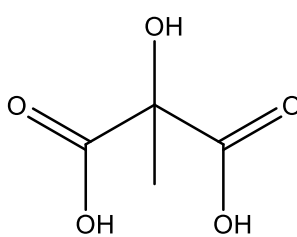
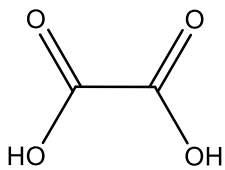
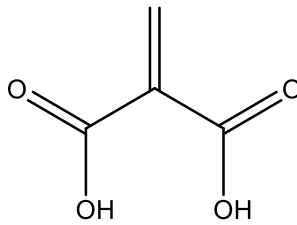
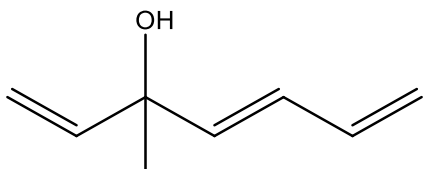
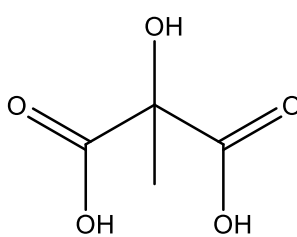
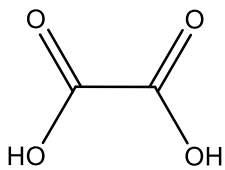
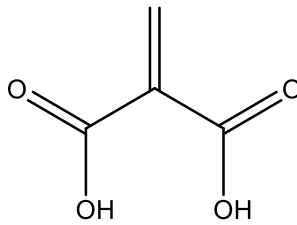
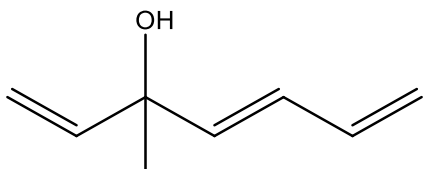
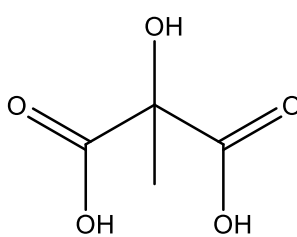
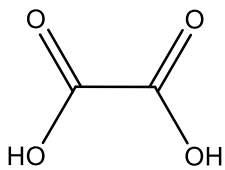
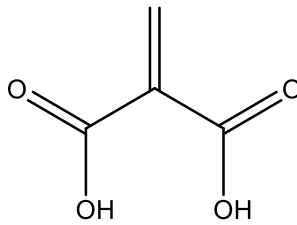
1	(a)	CO is able to bind strongly with haemoglobin in blood to form carboxyhaemoglobin, which prevents the bound haemoglobin from binding with oxygen. It also displaces oxygen bound to haemoglobin in a ligand exchange reaction. This will reduce the blood's capacity for transporting oxygen to the rest of the body.	[2]
	(b)	 dative bonds	[2]
	(c)	(i) $\text{C}_8\text{H}_{18} + \frac{23}{2}\text{O}_2 \rightarrow 6\text{CO}_2 + 2\text{CO} + 9\text{H}_2\text{O}$	[1]
		(ii)  * chiral centre (Other answers are accepted)	[1]
	(d)	Transition element complex often have an incompletely filled 3d subshell. In the presence of ligands, the 3d orbitals are split into 2 sets of non-degenerate orbitals with different energies. The difference in energies (ΔE) between these 2 sets of 3d orbitals is small and radiation from the visible region of the electromagnetic spectrum is absorbed when an electron moves from a lower energy d-orbital to another unfilled/partially-filled d orbital of higher energy. The colour observed correspond to the complement of the absorbed colours.	[3]
	(e)	(i) Homogeneous catalysis. All catalysts are in the same phase as the reactants (dissolved in methanol)	[1]
		(ii) $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$	[1]
		(iii) Step 3 and 5	[1]
		(iv) A and B: Catalyst. They are regenerated at Step 6 and 5 respectively in the reaction sequence. C: Reactant. It is consumed in step 4 and not regenerated at any point.	[3]

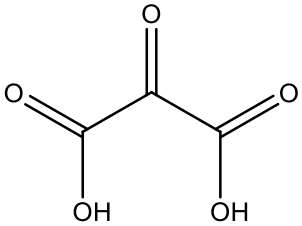
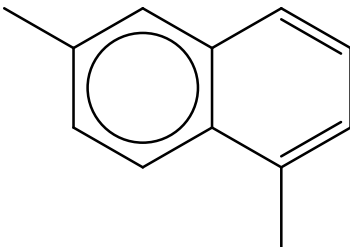
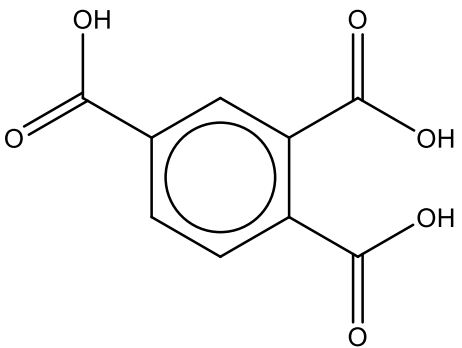
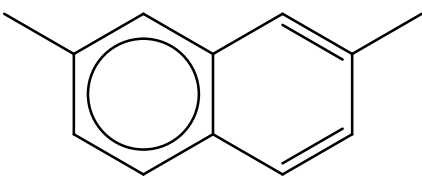
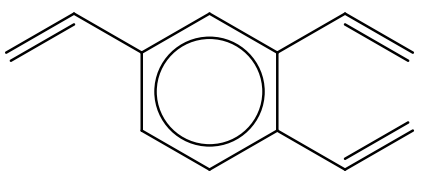
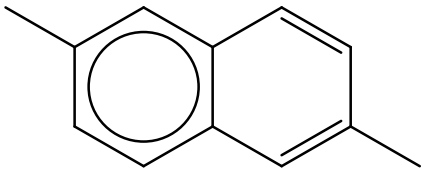
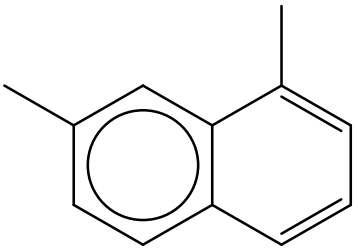
	(f)	(i)	$K_p = \frac{(P_{CH_4})(P_{H_2O})}{(P_{CO})(P_{H_2})^3}$ Units: atm ⁻² / Pa ⁻²	[2]
		(ii)	Partial pressure of a gas = mole fraction of gas × total pressure Equilibrium partial pressure of CH ₄ = Equilibrium partial pressure of H ₂ O = 0.12 × 32 = <u>3.84 atm</u> Equilibrium partial pressure of CO = 0.19 × 32 = <u>6.08 atm</u> Since a 1:3 molar ratio of CO and H ₂ is used, and 1 mol of CO reacts with 3 mol of H ₂ , Equilibrium partial pressure of H ₂ = 3(6.08) = <u>18.2 atm</u>	[2]
		(iii)	$K_p = \frac{(3.84)(3.84)}{(6.08)(18.2)^3} = 0.000400 \text{ atm}^{-2}$	[1]

2	(a)	• KI(aq) is added to Cl₂(aq) The solution pale yellow solution turns <u>brown</u>. $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq}) \quad E^\ominus = +1.36 \text{ V}$ $\text{I}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{I}^-(\text{aq}) \quad E^\ominus = +0.54 \text{ V}$ $E^\ominus_{\text{cell}} = +1.36 - (+0.54) = +0.82 \text{ V} > 0$ Equation : $\text{Cl}_2(\text{g}) + 2\text{KI}(\text{aq}) \rightarrow 2\text{KCl}(\text{aq}) + \text{I}_2(\text{aq})$ • KCl(aq) is added to Br₂(aq) No reaction. The <u>orange</u> solution remains. $\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-(\text{aq}) \quad E^\ominus = +1.36 \text{ V}$ $\text{Br}_2(\text{l}) + 2\text{e}^- \rightleftharpoons 2\text{Br}^-(\text{aq}) \quad E^\ominus = +1.07 \text{ V}$ For the oxidation of Cl⁻(aq) by Br₂(aq), $E^\ominus_{\text{cell}} = +1.07 - (+1.36) = -0.29 \text{ V} < 0$, reaction is not spontaneous. • KBr(aq) is added to I₂(aq) The <u>brown</u> solution remains unchanged. $\text{Br}_2(\text{l}) + 2\text{e}^- \rightleftharpoons 2\text{Br}^-(\text{aq}) \quad E^\ominus = +1.07 \text{ V}$ $\text{I}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{I}^-(\text{aq}) \quad E^\ominus = +0.54 \text{ V}$ For the oxidation of Br⁻(aq) by I₂(aq), $E^\ominus_{\text{cell}} = +0.54 - (+1.07) = -0.53 \text{ V} < 0$, reaction is not spontaneous	[4]
(b)	(i)	$\begin{array}{ccccc} \text{Br}_2(\text{l}) & + & 3\text{F}_2(\text{g}) & \xrightarrow{2(-301)} & 2\text{BrF}_3(\text{l}) \\ \downarrow +31 & & \downarrow & & \swarrow 2(44) \\ \text{Br}_2(\text{g}) & + & & & 2\text{BrF}_3(\text{g}) \\ \downarrow +193 & & \downarrow 3(158) & & \swarrow 6 \text{E}(\text{Br-F}) \\ 2\text{Br}(\text{g}) & + & 6\text{F}(\text{g}) & & \end{array}$ By Hess's law, $2(-301) = [31+193+3(158)] - [2(44)+ 6\text{E}(\text{Br-F})]$	[4]

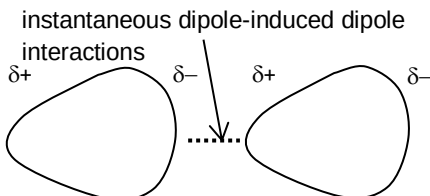
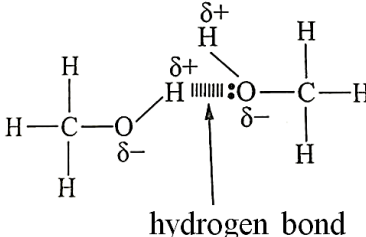
			$E(\text{Br}-\text{F}) = +202 \text{ kJ mol}^{-1}$	
		(ii)	$\Delta G_f^\ominus = \Delta H_f^\ominus - T\Delta S_f^\ominus$ $-241 = -301 - (298)(\Delta S_f^\ominus)$ $\Delta S_f^\ominus = -0.201 \text{ kJ mol}^{-1} \text{ K}^{-1}$ $\Delta S_f^\ominus < 0$ as there is a decrease in the number of moles of gaseous particles, resulting in a decrease in the disorder in the system.	[2]
	(c)	(i)	Structure of D :  Structure of E : 	[2]
		(ii)	Mechanism for reaction 1 : Electrophilic substitution Mechanism for reaction 2 : Free radical substitution	[2]
		(iii)	For the synthesis of 4-bromoethylbenzene: Generation of electrophile: $\text{Br}_2 + \text{AlCl}_3 \rightarrow \text{Br}^+ + \text{AlBrCl}_3^-$ 	[3]
		(iv)	Compound E would be a suitable starting material. 	[3]

3	(a)	(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$	[1]
		(ii)	$\text{Magnitude of lattice energy (LE)} \propto \frac{ Z^+ \times Z^- }{r^+ + r^-}$ MnO₂ has a more negative lattice energy. O²⁻ has the same charge as S²⁻ and is a smaller anion/has a smaller ionic radius.	[1]
	(b)	(i)		[1]
		(ii)	For Mn: oxidation number decreases from +4 in Mn ⁴⁺ to +2 in Mn ²⁺ . (Reduction) For S: oxidation number increases from -2 in S ²⁻ to -1 in S ₂ ²⁻ . (Oxidation)	[1]
	(c)	(i)	$8H^+ + MnO_4^- + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O \quad \times 2$ $Sn^{2+} \rightleftharpoons Sn^{4+} + 2e^- \quad \times 5$ Overall: $16H^+ + 2MnO_4^- + 5Sn^{2+} \rightarrow 2Mn^{2+} + 8H_2O + 5Sn^{4+}$	[1]
		(ii)	$E^\ominus / V$ $8H^+ + MnO_4^- + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O \quad +1.52$ $Pb^{2+} \rightleftharpoons Pb^{4+} + 2e^- \quad +1.69$ $E^\ominus_{\text{cell}} = 1.52 - 1.69 = -0.17 \text{ V} < 0$ (non spontaneous) Since reaction is non spontaneous, MnO₄⁻ does not oxidise Pb²⁺.	[1]
		(iii)	The solution in the conical flask changes from colourless to pale pink,	[1]

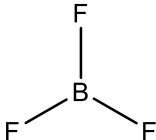
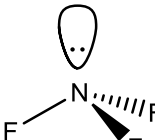
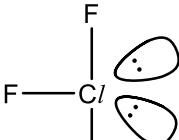
			which indicates that endpoint is reached.									
		(iv)	Amt of MnO_4^- used = $\frac{22.50}{1000} \times 0.0200 = 4.50 \times 10^{-4}$ mol Amt of Sn^{2+} in $25.0 \text{ cm}^3 = \frac{5}{2} \times 4.50 \times 10^{-4} = 1.125 \times 10^{-3}$ mol Amt of Sn^{2+} in $250 \text{ cm}^3 = 1.125 \times 10^{-3} \times 250/25 = 1.125 \times 10^{-2}$ mol % mass of Sn in solder = $\frac{0.01125 \times 118.7}{3.00} \times 100\% = 44.5\%$	[3]								
	(d)	(i)	F undergoes acid-metal reaction with Na metal which results in effervescence due to H_2 gas produced. Hence, F is an alcohol as the molecular formula has 1 O. F is not oxidised by warm acidified potassium dichromate. F is a <u>tertiary alcohol</u>.	[1]								
		(ii)	<table><tr><td>F</td><td></td></tr><tr><td>G</td><td></td></tr><tr><td>H</td><td></td></tr><tr><td>I</td><td></td></tr></table>	F		G		H		I		[6]
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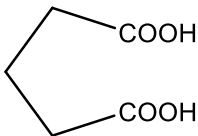
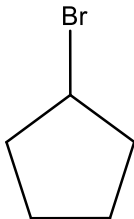
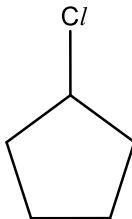
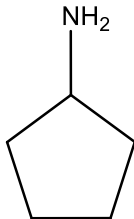
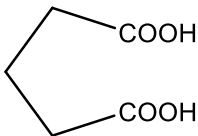
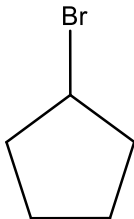
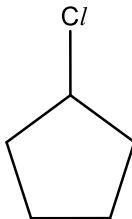
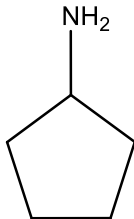
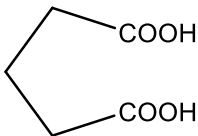
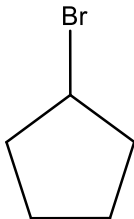
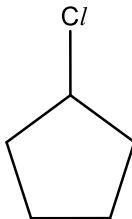
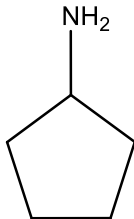
			J 		
			K 		
			L 		
		(iii) Cis-trans isomerism and enantiomerism Total no of isomers $= 2^2 = 4$		[2]	
		(iv) Structural isomerism    			

			Any 2	
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4	(a)	Both ethane and methane have simple covalent structures. The electron clouds of the methane and ethane molecules can be polarised at an instant in time due to unequal distribution of electrons, resulting in temporary dipole moments. These temporary dipoles induce a temporary dipole on neighbouring molecules and electrostatic forces of attraction between the molecules result in instantaneous dipole-induced dipole (id-id) interactions between the methane molecules, as well as between ethane molecules.  As ethane has more electrons, more energy is needed to overcome the stronger id-id interactions between ethane molecules compared to those between methane molecules. Thus, ethane has a higher boiling point than methane. Methanol also has a simple covalent structure. Hydrogen bonding can be formed between methanol molecules as shown as methanol has a hydrogen bonded to a strongly electronegative oxygen atom.  More energy is needed to overcome the stronger hydrogen bonding between methanol molecules compared to the weaker id-id interactions between ethane molecules. Thus, methanol has a higher boiling point than ethane.	[6]	
	(b)	(i)	Alkanes are unreactive due to their strong C–C and C–H covalent bonds. Alkanes are also non-polar and do not react with polar reagents.	[2]
		(ii)	<u>With oxygen</u> Conditions: Heat Type of reaction: Combustion Equation: $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$	[4]

		<u>With chlorine</u> Conditions: Limited Cl_2 and UV light Type of reaction: Free radical substitution Equation: $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$	
(c)	(i)	$\text{CO}_2 + 8\text{H}^+ + 8\text{e}^- \rightleftharpoons \text{CH}_4 + 2\text{H}_2\text{O} \quad E^\ominus = +0.17 \text{ V}$ $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O} \quad E^\ominus = +1.23 \text{ V}$ $E^\ominus_{\text{cell}} = (+1.23) - (+0.17)$ $= \underline{\underline{+1.06 \text{ V}}}$	[1]
	(ii)		[3]
(d)	(i)	$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu} \quad E^\ominus = +0.34 \text{ V}$ $\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn} \quad E^\ominus = -0.76 \text{ V}$ $E^\ominus_{\text{cell}} = (+0.34) - (-0.76)$ $= +1.10 \text{ V}$ As the setup consists of 5 pairs of Cu and Zn discs, the voltmeter reading would be $5 \times 1.10 = 5.50 \text{ V}$.	[1]
	(ii)	Overall reaction: $\text{Cu}^{2+} + \text{Zn} \rightarrow \text{Cu} + \text{Zn}^{2+}$ When the current is drawn, Cu^{2+} is reduced to Cu and Zn is oxidised to Zn^{2+}. Thus, the mass of Cu discs will increase and the mass of Zn discs will decrease. As the reaction proceeds, $[\text{Cu}^{2+}]$ decreases and this causes the equilibrium position of $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$ to shift to the left, thus $E(\text{Cu}^{2+}/\text{Cu})$ becomes less positive. At the same time, $[\text{Zn}^{2+}]$ increases and this causes the equilibrium position of $\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$ to shift to the right, thus $E(\text{Zn}^{2+}/\text{Zn})$ becomes more positive. Since $E_{\text{cell}} = E(\text{Cu}^{2+}/\text{Cu}) - E(\text{Zn}^{2+}/\text{Zn})$, the voltage decreases.	[3]

5	(a)	(i)	VSEPR Theory is used to predict the shapes of molecules and or polyatomic ions and applies to the outer shell of the central atom. The electron pairs around the central atoms are arranged as far apart as possible to minimise inter-electron repulsions. Lone pair-lone pair (lp-lp) repulsions are greater than bond pair-lone pair repulsions (bp-lp), which are greater than bond pair-bond pair (bp-bp) repulsions.				[2]
		(ii)		BF ₃ trigonal planar 	NF ₃ trigonal pyramidal 	ClF ₃ T-shape 	[4]
			Reason	3 bp and no lp around B atom placed at 120° to minimise repulsion.	3 bp and 1 lp around N atom: bp-lp repulsion more significant than bp-bp repulsion, causing bp to be closer to each other.	3 bp and 2 lp around Cl atom: lp placed at equatorial position to minimise total inter-electron repulsion.	
	(b)	(i)	Treat the alcohol with PCl ₅ / PCl ₃ / SOCl ₂ or Heat the alcohol with HCl (g)				[1]
		(ii)	In increasing reactivity: chlorobenzene < chloroethane < ethanoyl chloride Both ethanoyl chloride and chloroethane undergo hydrolysis with water and with NaOH(aq). With water, ethanoyl chloride gives ethanoic acid and HCl while chloroethane gives ethanol and HCl. With NaOH(aq), ethanoyl chloride produces sodium ethanoate and sodium chloride, while chloroethane produces ethanol and sodium chloride. However, <u>ethanoyl chloride reacts with water and NaOH(aq) in the cold, while chloroethane only reacts with water and NaOH(aq) when heated.</u> <u>Chlorobenzene does not react</u> even when heated with water or NaOH(aq). Ethanoyl chloride is the most reactive as the <u>carbonyl carbon atom in –COCl is highly electron-deficient as it is bonded to 2 electronegative atoms, O and Cl, and hence is most susceptible to attacks by nucleophiles like water and OH⁻.</u> Chlorobenzene is unreactive due to the <u>partial double bond character of</u>				[4]

			the C–Cl bond due to the delocalisation of the lone pair on Cl atom into the benzene ring, making the C–Cl bond stronger.							
	(c)	(i)	In an aqueous solution or in an organic solvent like CCl ₄ .	[1]						
		(ii)	A chlorinated phenol is a stronger acid than phenol. In the chlorinated phenoxide ion, the electron-withdrawing chlorine atoms increase the delocalisation of the negative charge into the benzene ring, making the chlorinated phenoxide ion more stable than the phenoxide ion.	[1]						
	(d)	<u>Reagents and conditions</u> Step 1: acidified KMnO₄(aq), heat under reflux Step 2: excess SOCl₂ or PCl₅ or PCl₃ Step 3: HBr(g) or HCl(g), rtp Step 4: excess ethanolic NH₃, heat in a sealed tube <u>Structures</u> <table border="1"> <tr> <td>  </td> <td>   OR </td> <td>  </td> </tr> <tr> <td>M</td> <td>N</td> <td>P</td> </tr> </table>				  OR		M	N	P
	  OR									
M	N	P								

[7]